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# Take Bitcoin into your portfolio: a novel ensemble portfolio optimization framework for broad commodity assets

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The emergence and growing popularity of Bitcoins have attracted the attention of the financial world. However, few empirical studies have considered the inclusion of the newly emerged commodity asset in the global commodity market. It is of great importance for investors and policymakers to take advantage of this asset and its potential benefits by incorporating it as a part of the broad commodity trading portfolio. In this study, we propose a novel ensemble portfolio optimization (NEPO) framework utilized for broad commodity assets, which integrates a hybrid variational mode decomposition–bidirectional long short-term memory deep learning model for future returns forecast and a reinforcement learning-based model for optimizing the asset weight allocation. Our empirical results indicate that the NEPO framework could effectively improve the prediction accuracy and trend prediction ability across various commodity assets from different sectors. In addition, it could effectively incorporate Bitcoins into the asset pool and achieve better financial performance compared to traditional asset allocation strategies, commodity funds, and indices.

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## Introduction

It is important to consider the distributions and classes of different assets in investor portfolios in maximizing the asset returns for a given level of volatility (Konno and Yamazaki [1991](#)). Based on the belief that distinctive factors influence the price and risk movements of different assets and that correlation among global financial products is relatively low, investors construct their portfolios by diversifying investment in various assets.

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investors to embrace this cryptocurrency as an investment option. For example, Bitcoin's price resurgence in 2021 was partly fueled by the Wall Street billionaires who publicly supported and invested in the asset.<sup>1</sup> However, due to its high volatility, most investors are hesitant to solely invest in this asset. Instead, many trading firms seek to incorporate it into their portfolio along with other traditional commodities to hedge against its potential volatility risks (Liu and Tsyvinski [2018](#)).

Bitcoin's commodity properties have been investigated by some studies. For example, using a conditional correlation model, Bouri et al. ([2017](#)) suggest that Bitcoin can serve as a safe haven for other major commodities in the global commodity market system. Selmi et al. ([2018](#)) use the quantile regression to investigate the economic characteristics of Bitcoin, indicating that it is both a hedge and safe haven for oil price movements. As a result, Bitcoin is considered the "new gold" for its safe haven properties, which are similar to that of gold and serve as a potential hedge or safe haven asset for financial portfolio optimization (Selmi et

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With the development of computer technologies, deep learning prediction models have been introduced to improve the financial time series data forecasting accuracy. Compared to traditional techniques based on econometrics and machine learning techniques that are unable to perform well on forecasting multivariate time series data due to noise disturbances (Altan et al. [2019](#); Galankashi et al. [2020](#); Jalali and Heidari [2020](#)), deep learning techniques are observed to be more effective. For example, Atsalakis et al. ([2019](#)) introduce a novel Neuro-fuzzy technique with artificial neural networks (ANN) to forecast the market trends in cryptocurrency prices, which show an improvement in prediction accuracy compared to traditional prediction methods. Dutta et al. ([2020](#)) employ a gated recurring unit model to predict the price movements of Bitcoins and achieve a better forecasting performance. Long et al. ([2019](#)) propose a multi-filter neural network for stock price prediction. Further, Li et al. ([2019](#)) develop a crude oil price prediction system based on the convolutional neural networks. Among all the deep learning techniques, the recurrent neural network (RNN) models have displayed a superior performance over others in terms of time series prediction

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Numerous studies have employed this hybrid forecasting approach and demonstrated its effectiveness in improving the time series forecasting performance (Yu et al. [2015](#); Wen et al. [2017](#)). Despite the improved prediction performance, there might be a potential problem with the decomposition and ensemble approach. Estimation errors generated while forecasting the individual sub-modes tend to accumulate during the aggregation process. This accumulation may cause significant discrepancies between the actual and predicted values, which could negatively affect the prediction performance (Zhu et al. [2019](#)). Therefore, we propose a new hybrid deep learning-based forecasting approach in our portfolio optimization framework to mitigate this problem. Unlike previous studies, our forecasting approach eliminates the aggregation step by directly generating the final prediction results using all the intrinsic modes as inputs simultaneously, which can potentially reduce the aggregation errors.

The literature has adopted many portfolio models, such as the cardinality constrained model

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choose. However, the portfolio weight allocation is a continuous action space in reality, as each asset can be potentially given any weight between 0 and 100%. Therefore, although the portfolio performances have improved, the previously implemented model may have ignored the possibility of other allocations that do not exist in the pre-determined action space. To address this limitation and improve the model's consistency, we develop a portfolio optimization framework based on a deep deterministic policy gradient model (Lillicrap et al. [2015](#)) that can effectively consider the continuous characteristics of the weight allocation action space. In addition, the previously employed portfolio optimization models typically allocate asset weights to optimize the portfolio performance directly based on the forecasted trends of the assets without considering the potential prediction errors. Hence, these models assume that the forecasted value is entirely accurate, which is unrealistic in practice. Therefore, our portfolio optimization framework attempts to address this issue by considering the prediction errors that may arise in the forecasting process, which can potentially allocate weights more effectively to improve the portfolio performance.

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empirical results suggest that the proposed VMD-BiLSTM model can effectively improve the prediction accuracy and trend prediction ability across various commodity assets.

We compare the performance of our portfolio optimizer to that of other commodity funds, indices, and asset allocation strategies in terms of an annualized return and Sharpe ratio. The empirical results indicate that our ensemble portfolio optimization framework can generate higher returns and a better Sharpe ratio than the others. The results also indicate that by including Bitcoins in the commodity portfolio, asset managers can achieve higher returns without being exposed to significant financial risks. We find that it is possible to take advantage of the returns generated from Bitcoins while reducing the investment risks caused by its extreme volatilities. Overall, employing the proposed ensemble portfolio optimization framework and considering Bitcoin a traditional commodity portfolio can generate better fund performances for asset management and portfolio profits for commodity investors.

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portfolio weights based on the predicted returns and forecasting evaluation metrics. The detailed framework is shown in Fig. [1](#).

**Fig. 1**

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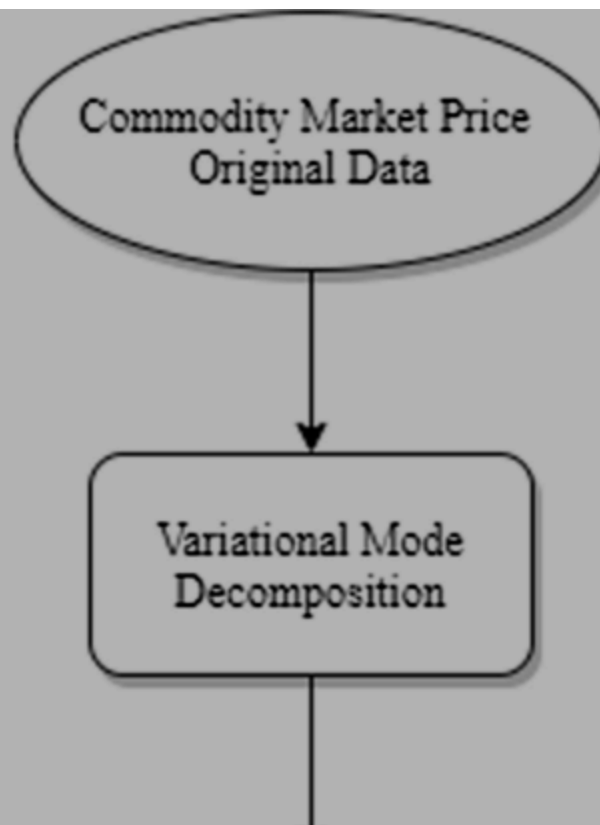
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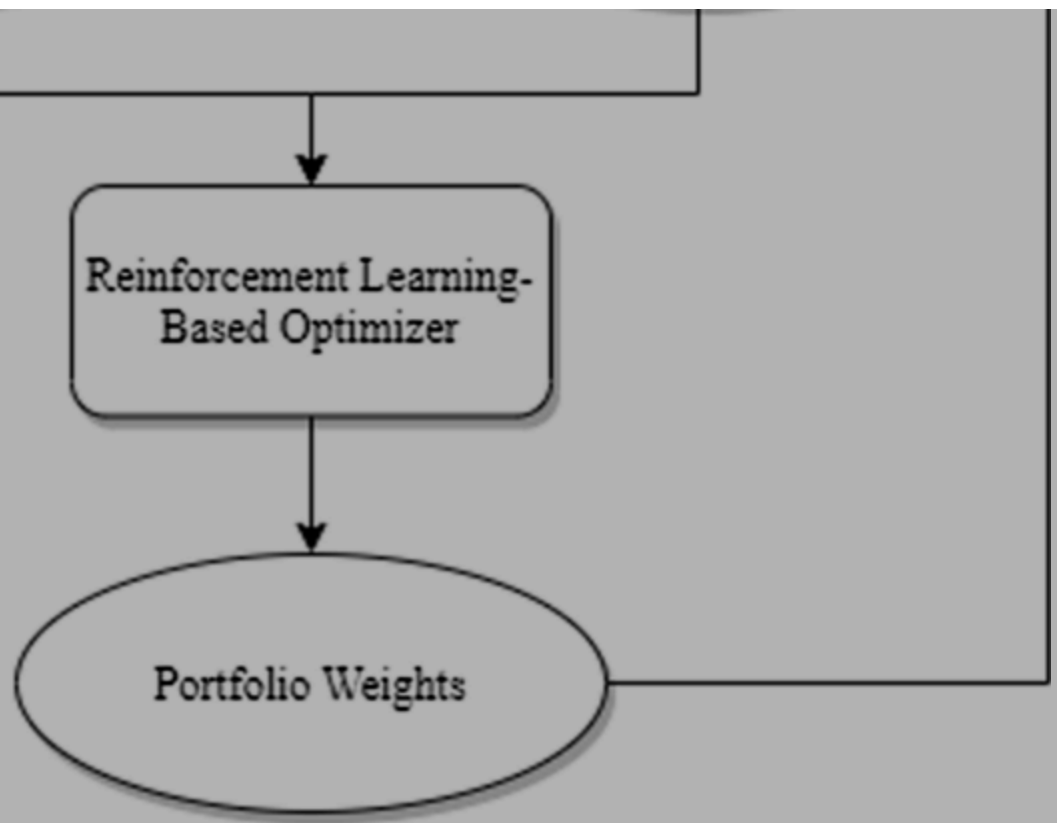
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estimated center frequency.

*Step 3:* Obtain the bandwidth of each mode  $(u_k)$  by conducting the  $(H^1)$  Gaussian smoothness on the demodulated signal.

The iterative minimization process seeks to minimize the total bandwidth of each mode, which can be expressed in the following form:

$$\min_{\{u_k\}, \{\omega_k\}} \left\{ \sum_{k=1}^K \partial_t \left[ \left( \delta(t) + \frac{j}{\pi t} \right) \otimes u_k(t) \right] e^{j(\omega_k t)} \right\}^2 \quad \text{s.t.} \quad \sum_{k=1}^K u_k = f(t), \quad (1)$$

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The optimal solution is obtained using the alternative direction method of multipliers (Hestenes [1969](#)), while the original input signal  $(f(t))$  is decomposed into  $(K)$  sub-signal modes.

## BiLSTM neural networks

The bidirectional RNN was proposed by Schuster and Paliwal ([1997](#)). It utilizes both forward and backward information in the data. As illustrated in Fig. [2](#), the bidirectional RNN structure contains two unidirectional hidden layers, where one layer processes information from the forward direction and the other from the backward direction. The forward and backward unidirectional layers are concatenated to one output layer, such that the neural networks can extract bidirectional sequential relationships in the time series data. Compared to traditional unidirectional neural networks, it can preserve information from both the past and future.

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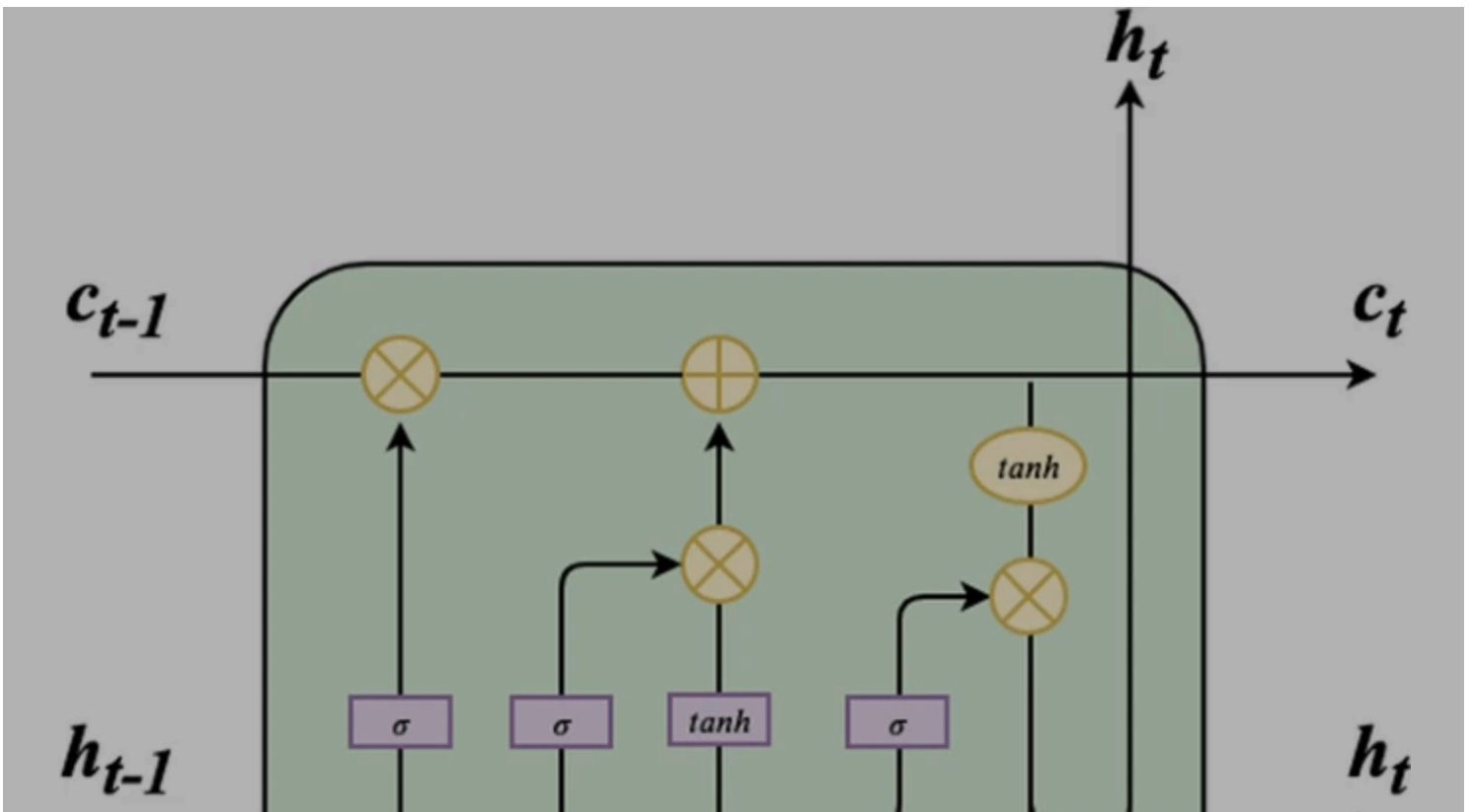
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(4)

A  $\tanh$  layer is utilized to generate a new memory cell block  $\tilde{C}_t$ . The existing memory cell block  $C_t$  is then updated, while the output gate  $o_t$  and hidden state  $h_t$  are generated:

$$i_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

(5)

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

(6)

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$$\pi : S \times \{A\} \rightarrow \{0,1\}$$

(9)

$$\pi \left( \{a\}_s \right) = \Pr \left( \{a_t\} = \{a\}_s \mid s_t = s \right).$$

(10)

The state spaces contain all the possible allocation of portfolio weights, while the actions are the set of possible allocations from state spaces. At each time step  $(t)$ , the state  $(s_t)$  and action  $(a_t)$  can be expressed as follows:

$$a_t = w_t = (w_{1,t}, \dots, w_{n,t})$$

(11)

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utilized indicator and serves as the baseline of portfolio ratio improvements in academia and industry (Farinelli et al. [2008](#); Kapsos et al. [2014](#)).  $(Q_t)$  and  $(\widetilde{Q}_t)$  in Eq. (12) denote the weighted Sharpe ratio (Sharpe [1994](#)) of the newly allocated portfolio weights and the portfolio weights observed in previous studies, respectively.  $(\widetilde{RMSE}_t)$  and  $(RMSE_t)$  represent the weighted root mean squared error (RMSE) of the prediction models for the new portfolio weights and the weights computed in previous studies, respectively. The reinforcement learning model is trained to find a set of portfolio weight allocations that will maximize the expected return:

$$V_{\pi}^s = E \left[ \sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s \right]. \quad (14)$$

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We select five major commodity markets to construct our commodity portfolio, which consists of stocks, agriculture, energy, precious metal, and the newly emerged cryptocurrency commodities. In the portfolio, each market is represented by its leading commodity, which includes the S&P 500 stock index, wheat, WTI crude oil, gold index, and Bitcoin.

The data are obtained from Yahoo Finance, from which we download the daily closing price of the S&P 500 index, wheat, WTI crude oil, gold, and Bitcoin from January 2, 2013 to February 21, 2020, obtaining 1797 observations. As Bitcoin is traded continuously throughout the day, its opening price generally refers to the price at 12:01 AM UTC and the closing price to that at 11:59 PM UTC on any given day. A graphical representation of the data for each commodity is provided in Fig. [3](#).

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(right skewed). In addition, the augmented Dickey–Fuller test indicates that the time series data for the stock (SPY), crude oil (WTI), gold, and Bitcoin are all non-stationary and have a unit root. The null hypothesis of the augmented Dickey–Fuller test for wheat is not rejected at the 10% level of statistical significance, which means that the series is stationary.

Table 1 Descriptive statistics

Moreover, this study conducts a correlation analysis among the commodities. The Pearson correlation coefficients are shown in Fig. 4. The results suggest that there exist moderately weak non-linear relationships among the commodities. In particular, Bitcoin is negatively correlated with all the other commodities. In addition, there exists a strong positive

Table 1 Descriptive statistics (continued) (continued) (continued) (continued) (continued)

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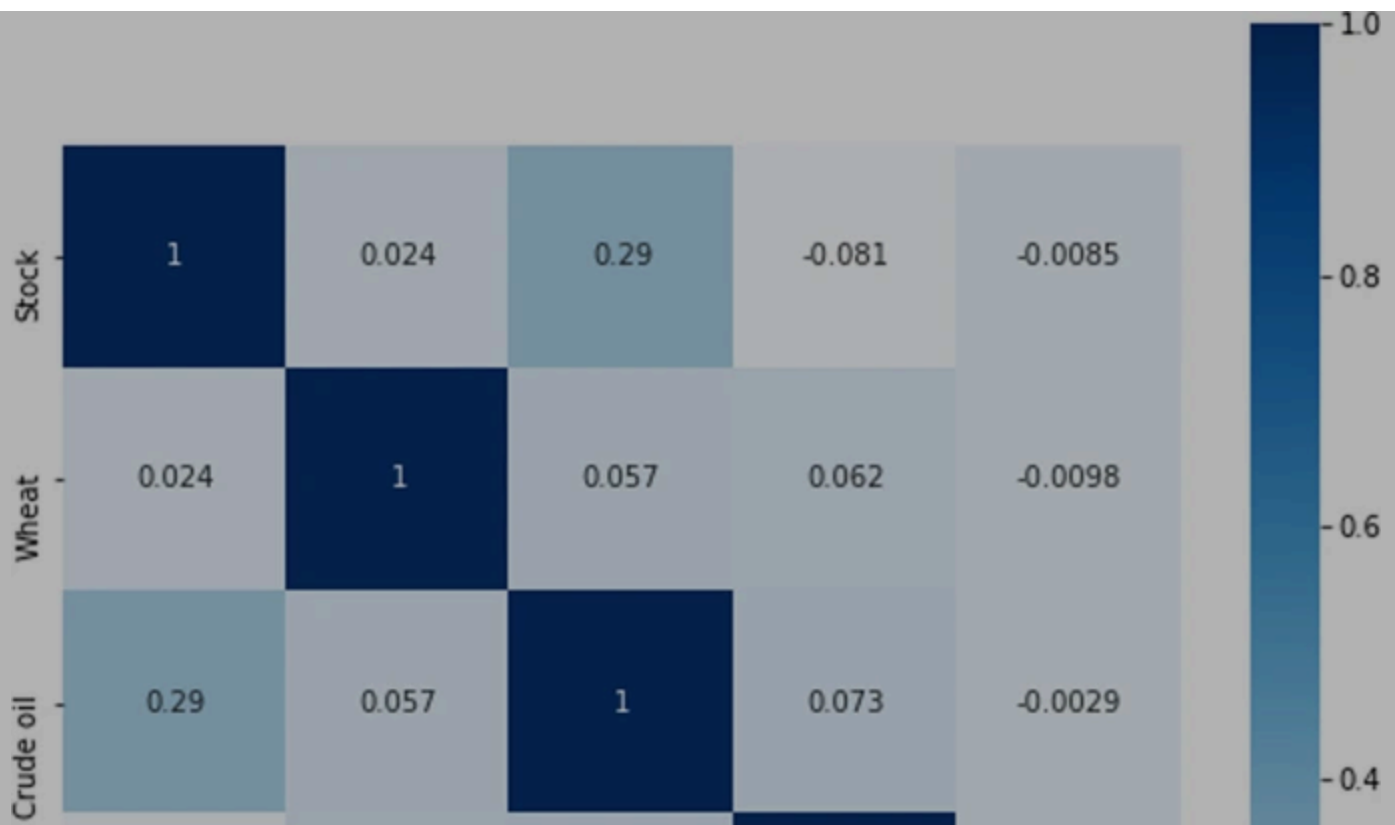
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consists of 1421 observations from January 30, 2013 to September 19, 2018, while the testing set consists of 356 observations from September 20, 2018 to February 21, 2020.

To eliminate the differences in the variable dimension and increase model forecasting reliability, we normalize the data in the range of [0,1] as shown below:

$$\widehat{x_t} = \frac{x_t - \min x_t}{\max x_t - \min x_t}, \quad (15)$$

where  $(x_t)$  denotes the true value of the time series at time  $(t)$ , while  $(\max x_t)$  and  $(\min x_t)$  are the maximum and the minimum true values of the time series, respectively.

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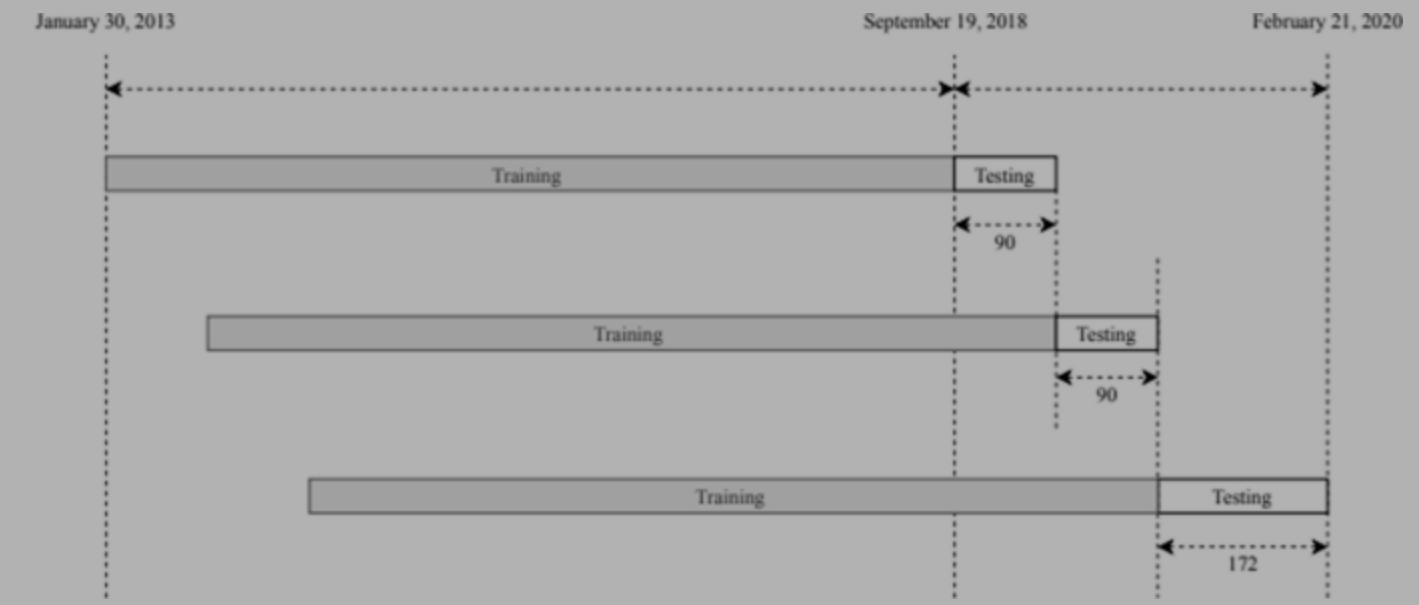
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Fig. 5



Rolling forecast process

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$$MAE = \frac{1}{N} \sum_{t=1}^N |x_t - \tilde{x}_t|, \quad (19)$$

(19)

where  $\tilde{x}_t$  and  $x_t$  ( $t = 1, 2, \dots, N$ ) are the predicted and actual true values at time  $t$ , while  $N$  represents the total number of data points in the testing set.

In addition, we introduce DA as the metric to assess the market trend predictive ability of the model:

$$DA = \frac{1}{N} \sum_{t=1}^N a_t, \quad (20)$$

(20)

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$$SR = \frac{r_p - r_f}{\sigma_p},$$

(22)

where  $(r_p)$  denotes the annualized return of the portfolio,  $(\sigma_p)$  is the annualized volatility of the portfolio, and  $(r_f)$  represents the nominal risk-free rate. As suggested in previous studies (Fabozzi et al. [2007](#); Ackerman et al. [2013](#)), we set the nominal risk-free rate  $(r_f = 2\%)$ .

We consider several other portfolios, including the equal-weighted portfolio (the five chosen commodities are allocated equal weights throughout the trading period) and non-Bitcoin portfolio (only SPY, wheat, WTI, and gold are considered), for comparing performance. We also obtain the financial performance data from similar exchange traded funds (ETF) for comparison, which includes the broad commodity ETF and Bitcoin ETF. The financial data

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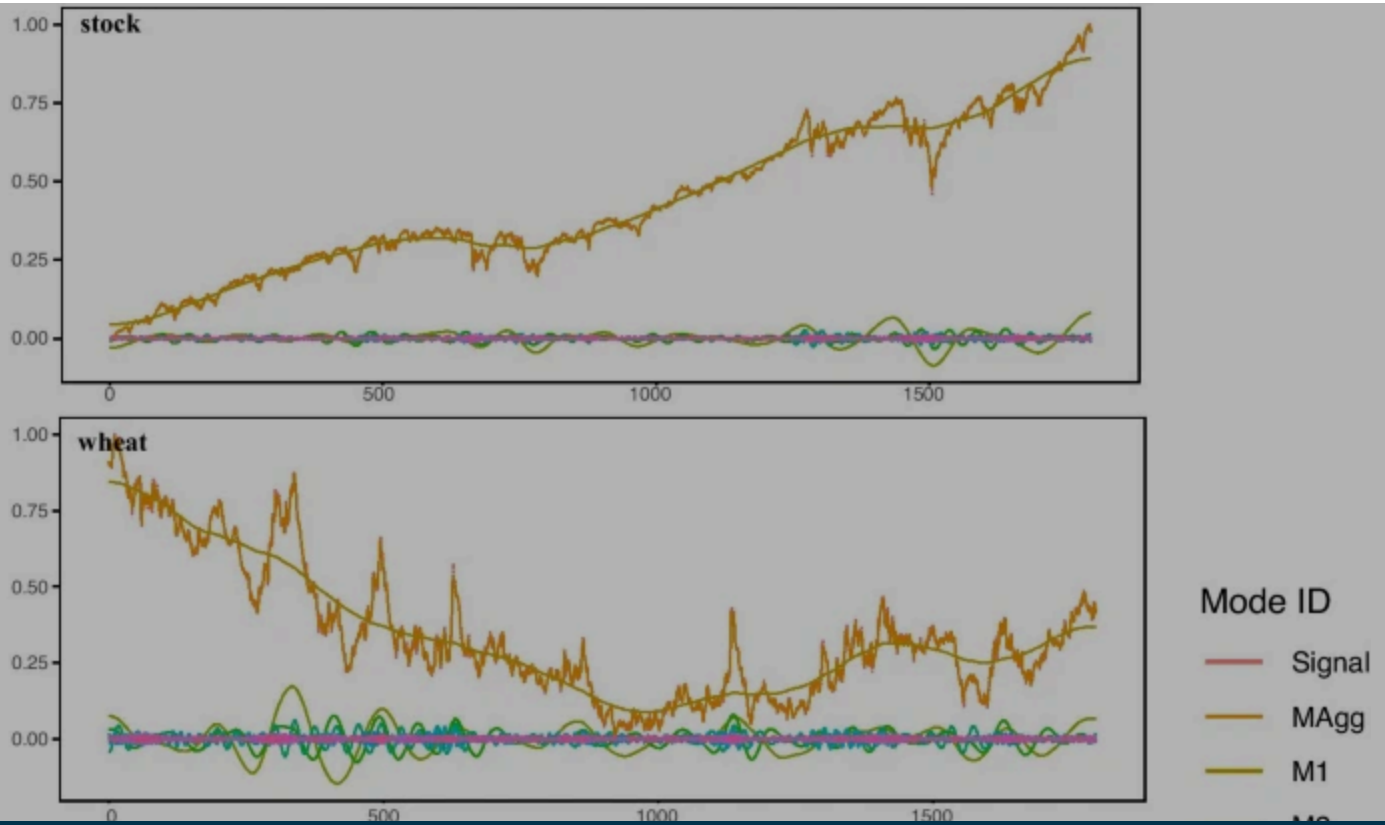
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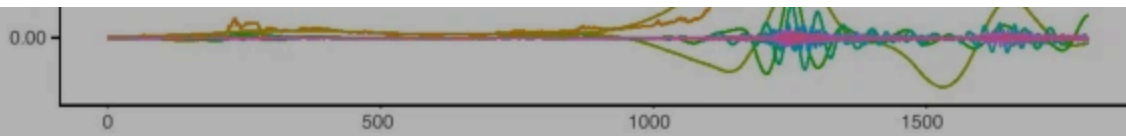
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Data decomposition results

As we can see from Fig. 6, the historical daily closing price data for each commodity are decomposed into 11 sub-series modes labeled from M1 to M11, respectively. For each commodity, their decomposed modes display different cyclicity and fluctuation patterns. The sub-series modes range from low frequency to high frequency. Specifically, the M1 modes have the lowest frequency, reflecting the long-term trends for the time series. However, M2 to M5 represent the medium frequency modes, which show the periodicity of the price fluctuation. Lastly, M6 to M11 are the high-frequency modes, which represent the short-term fluctuations in the data.

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**Table 2 Comparison of prediction performances for the proposed model and benchmarks**

After analyzing each commodity, we find that the VMD-BiLSTM model displays the most drastic improvements in RMSE and MAE performances in the case of the benchmark models. The VMD-BiLSTM model obtains the highest DA out of all the models. This superior performance indicates that our prediction model can effectively capture and forecast the movement trend for all the selected commodity markets. Moreover, the high prediction accuracies achieved by our VMD-BiLSTM model across all the commodities indicate that it is not overfitted to a particular dataset. Thus, it could be generalized for all commodity markets.

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### Table 3 Robustness test with different hyperparameters

The results in Table 3 indicate that the different hyperparameter combinations can yield varied model prediction accuracies, where the bold fonts present the prediction results of 11 dimension hidden layers and 0.01 LR settings for each market. It is clear that when the dimension of the hidden layers is set to 11 and LR is set to 0.01, our prediction model obtains the best prediction results. Further, the consistently superior results across all the commodities indicate that our model is not overfitted to a particular dataset.

To further verify the robustness of our prediction model, we use the same prediction model with the best hyperparameter settings on four different datasets that include the first 95%, 99%, 95%, 100% of the training data. The results are shown in Table 4. The results show that the prediction model with the best hyperparameter settings can achieve high accuracy across different datasets.

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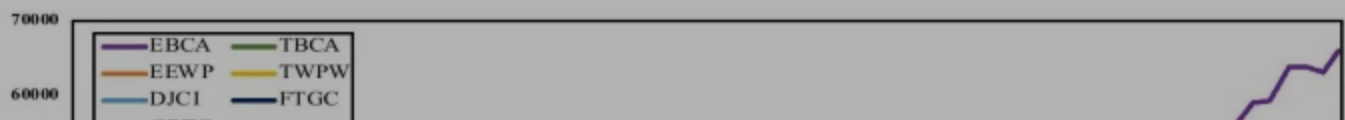
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every five days. After obtaining the allocation weights in each portfolio, we calculate the actual annualized returns, volatility, and Sharpe ratio of the portfolios using real-time commodity prices. To further verify the practicality of our strategy in the real world, we consider the transaction fees of each commodity, which are collected from Yahoo Finance. To evaluate the performance of each selected portfolio, we divide the entire trading interval into quarters (four months). In our trading policy and simulation, we ensure that our assets are sufficiently large to cover the trading volumes for all commodities. As a result, the initial investment capital for each trading strategy is set at \$10,000. The investment asset and indicator comparisons are separately illustrated in Fig. 7 and Table 5.

**Fig. 7**



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NEPO framework: the extended broad commodity asset (EBCA) portfolio and the traditional broad commodity asset (TBCA) portfolio. The EBCA portfolio contains all the selected commodity assets (SPY, wheat, WTI, gold, and Bitcoin). In contrast, the TBCA portfolio includes only the traditional commodity assets (SPY, wheat, WTI, and gold). To evaluate the performance of our portfolios, we compare the results to that of the portfolios constructed using other strategies, similar indices, and funds, such as extended equal-weighted portfolio, traditional equal-weighted portfolio without Bitcoin, Dow Jones Commodity Index (DJCI), top field broad commodity ETF (FTGC), and top field Bitcoin ETF (GBTC) that only consists of digital assets.

The results in Table 5 and Fig. 7 show that our portfolio constructed using the reinforcement learning model outperformed the other portfolio, indices, and funds in terms of financial performances for all the trading periods in the analysis. First, our constructed EBCA and TBCA portfolios are unique in that they can maintain consistent positive returns throughout

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113.91% in 03/2019–05/2019, while the portfolio volatility attains 103.04%. As a result, this “high risk, high reward” characteristic of Bitcoin exposes investors to significant risks. Compared with the GBTC funds, our EBCA portfolio increases the average returns by 78% from 19.48 to 34.67% and significantly reduces the portfolio risk from 90.12 to 24.27%. By incorporating Bitcoin as part of our constructed portfolio, we can take advantage of the attractive returns of Bitcoin while limiting the exposure to the risks of the asset, which may be a viable strategy for individual investors.

The results in Table [5](#) also show that our EBCA portfolio outperformed the TBCA traditional commodity portfolio in four of six intervals by achieving better Sharpe ratios. Overall, the EBCA portfolio obtains average financial returns of 20.08% with an average volatility of 23.09%. With the inclusion of Bitcoin as part of the portfolio, the average volatility of the portfolio throughout all the quarters increased by 5.1% to 24.27%. Despite the slight increase in portfolio risks, the average returns of the portfolio saw a significant jump to 34.67%,

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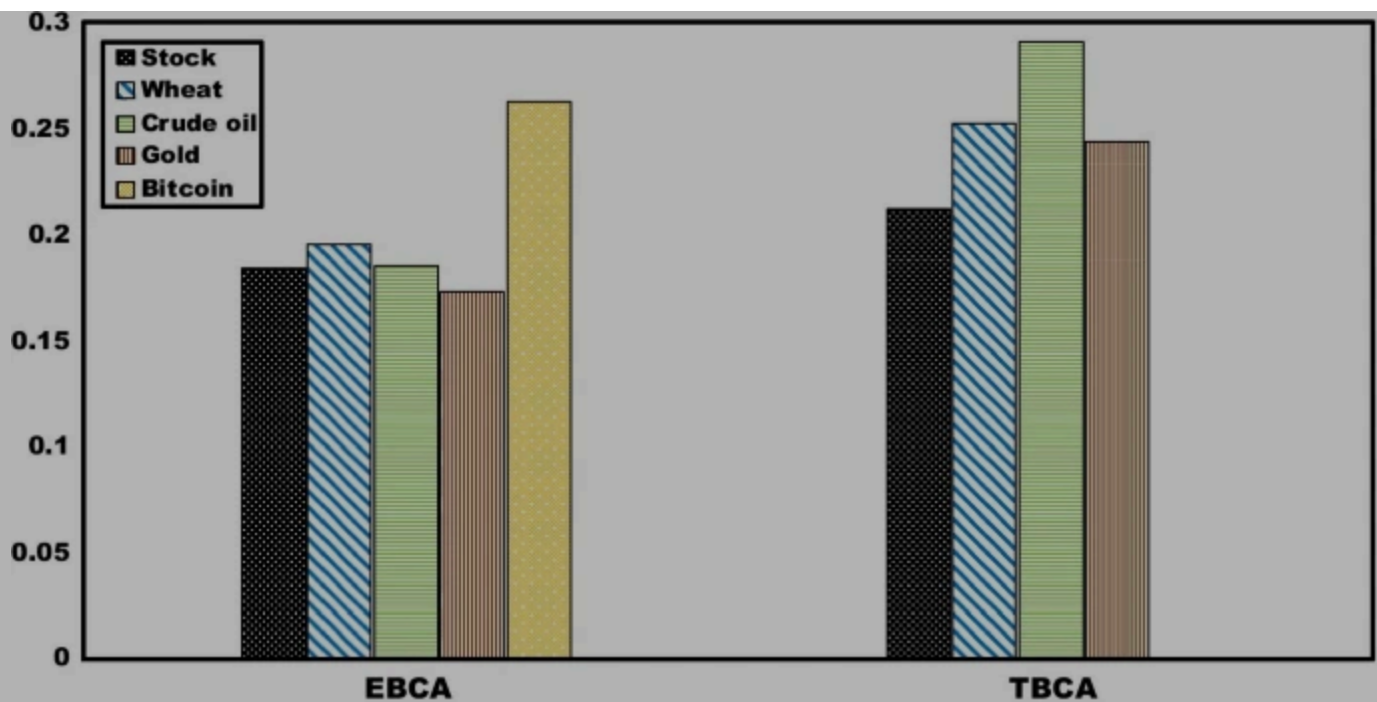
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Overall portfolio distribution of the EBKA and TBKA portfolios

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commodity funds, indices, and asset allocation strategies. In addition, by incorporating Bitcoin into the asset pool, our portfolio optimization framework can increase the financial performance of the broad commodity portfolio by taking advantage of its high returns and effectively reducing its inherent risks.

This study adds to the literature through multiple channels. First, our broad commodity portfolio optimization framework serves as an early attempt to incorporate Bitcoin in the asset pool. Further, it could be effectively used to increase the diversification premiums of the portfolio without greater exposure to investment risks. Our VMD-BiLSTM forecasting approach differs from other hybrid forecasting approaches applicable in financial time series analysis. It directly generates the forecasting results by simultaneously using all the extracted intrinsic modes as prediction model inputs. Our proposed model can effectively avoid the estimation errors that tend to accumulate in the current ensemble prediction approaches by eliminating the aggregation step. Finally, our proposed NEPO framework

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## Notes

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1. See <https://www.cnbc.com/2021/01/04/bitcoin-btc-rally-partly-driven-by-more-institutional-investors-pwc-says.html>.

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## Contributions

YL conceived of the presented idea, contribute to the methodologies and analyzed the data. SJ initiated the subject with YL, contributed to the review of literature and interpretation of the results. YW modified the manuscript and contribute to the discuss of the empirical results. SW suggested such a topic and supervised throughout the manuscript preparation. All authors read and approved the final manuscript.

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## Ethics declarations

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